

Armin Soleymani — Curriculum Vitae

Simon Fraser University – School of Computing Science

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🔗 ArminSLM

RESEARCH INTERESTS

- Computer Vision
- Computer Graphics
- Applied Deep Learning
- Robotics
- Generative AI
- Video Models

EDUCATION

- Master of Applied Science 2026–Present / Burnaby-Canada
 *Simon Fraser University*
Computing Science *Research: 3D Geometry Modeling and AI Video Models*
Visual Computing
- Bachelor of Science 2021–Present / Tehran-Iran
 *Amirkabir University of Technology*
Electrical Engineering *Last two years GPA: 3.6/4* *GPA: 3.21/4*
Telecommunications *Out of 68 credits* *Out of 140 credits*

PUBLICATIONS

- **FiReT: A Neural Radiance Fields Framework for Wireless Field Reconstruction and Transmitter Placement**
 - Extended NeRF to the RF wave propagation domain for reconstructing wireless channels from sparse measurements
 - Integrated ray-tracing and Direction-of-Arrival (DoA) guidance to improve convergence and physical consistency
 - Enabled transmitter placement optimization by ranking candidate locations through learned field representations
 - [Paper accepted for presentation at IEEE IKT 2025, indexed by IEEE Xplore](#)
- **RadarPose: Frequency-Domain Diffusion for mmWave Radar-Based Human Pose Estimation**
 - Design a diffusion-based framework for human pose estimation from mmWave radar signals
 - Applied frequency-domain diffusion to better capture global motion structure and reduce high-frequency noise.
 - Incorporated spatio-temporal features with diffusion models and GCN to improve joint localization under occlusion
 - Paper in Preparation

HONORS

- Ranked among **1%** in university entrance exam, among more than **142,000 participant**
- **1st Place** – RoboCup Salvador 2025, Rescue Line League – RoboCup Federation
- **1st Place** – RoboCup Sydney 2019, Rescue Line League – RoboCup Federation
- **2nd Place** – RoboCup Eindhoven 2024, Rescue Line League – RoboCup Federation
- **1st Place** – FIRA RoboWorld Cup Taichung City 2018 – Federation of International Robot-Sport Association
- Awarded a **Fellowship** for the B.Sc. program in Electrical Engineering at the Amirkabir University of Technology

WORK & RESEARCH EXPERIENCE

-  **Iran Telecommunication Research Center (ITRC)** July 2024 - October 2024
Deep Learning–Based Image Segmentation for Estimating Cultivated Areas of Strategic Crops: 
 - Developed a GAN-based augmentation pipeline to enrich satellite imagery datasets for segmentation tasks
 - Integrated real and synthetic images into a U-Net model for improved land segmentation
 - Refined segmentation outputs using Dense Conditional Random Fields (CRF) to boost spatial accuracy
-  **Microwave Measurement Research Lab** October 2024 - May 2026
 - Conducting research on Neural Radiance Fields (NeRF), Signal Processing and Computer Vision methods for radio signal processing, wireless channel estimation, RF propagation modeling and microwave measurement
-  **Benis Food Industries Company** Feb 2024 - Feb 2026
Machine Vision and Automation developer :
 - Designed an vision system to detect and classify objects on the cake packaging line
 - Integrated object detection with Festo pneumatic actuators to enable precise separation and automation
-  **RoboCup Federation** July 2024 - May 2026
Rescue Line League Technical Committee member:
 -  *RoboCup 2025, Salvador*
 -  *RoboCup 2024, Eindhoven*
 -  *RoboCup IranOpen 2025, Tehran*
 -  *RoboCup IranOpen 2024, Tehran*

- o  **RoboDanesh Academy**
Robotics Instructor and Curriculum Developer:
 - Python programming for image processing
 - C++ programming using AVR controllers and Arduino

June 2022 - May 2026

SELECTED PROJECTS

- o **Autonomous Driving with Vision-Based PPO Reinforcement Learning** Computational Intelligence
 - Trained a PPO-based reinforcement learning agent in Webots to autonomously drive and avoid obstacles 
 - Achieved stable lane-keeping behavior using OpenCV for visual lane detection and object recognition
 - Integrated sonar sensor readings with vision data to enhance perception and design the reward function
 - Developed a modular simulation pipeline for testing autonomous policies in dynamic environments
- o **Hybrid ResNet-Inception CNN Classifier for Tiny ImageNet Dataset** Multimedia Systems
 - Designed a custom CNN architecture integrating concepts from ResNet and Inception models 
 - Applied the model to the Tiny ImageNet dataset for multi-class image classification
 - Used multiple convolutional branches with different kernel sizes to capture multi-scale features
- o **Twitter Emotion Classifier using Transformer Encoder** Multimedia Systems
 - Developed a transformer-based model from scratch in PyTorch to classify emotions in tweets 
 - Preprocessed raw Twitter data using custom tokenization, padding, and truncation logic
 - Engineered a Transformer Encoder with positional encoding to retain sequence semantics
 - Captured contextual dependencies effectively for emotion prediction in short-text inputs
- o **Speech Emotion Recognition using Wav2Vec2 and Text-Based Transformer** Multimedia Systems
 - Leveraged Wav2Vec2 from HuggingFace to transcribe emotional audio into structured text sequences 
 - Unified emotion labels across multiple datasets to enable consistent and scalable model training
 - Developed a text-based Transformer classifier for emotion prediction from transcribed speech
- o **Interpretable MNIST Classification with Saliency Maps and FGSM Attacks** Multimedia Systems
 - Built a CNN-based image classifier for MNIST with a focus on visual interpretability and robustness 
 - Developed custom saliency mapping tools to expose model attention over input pixels
 - Highlighted key features via gradient-based methods to enhance decision transparency
 - Evaluated model stability under FGSM adversarial attacks to reveal vulnerability patterns
- o **ECG Signal Processing with RNNs for Denoising and R-Peak Detection** Multimedia Systems
 - Designed a pipeline to preprocess and augment ECG signals with Gaussian noise and temporal segmentation 
 - Implemented an LSTM model for sequence-to-sequence denoising, evaluated with MAE and RMSE
 - Developed a GRU-based classifier to detect R-peak presence in ECG segments through binary labeling
 - Automated CSV export of predictions and visualized denoised signals, error curves, and detection outcomes

LANGUAGE SKILLS

- English: Fluent
- Persian: Native
- Azari: Conversational

COURSES

- | | | | |
|--|----------|---|-------------------------|
| o  Computational Intelligence | 4/4 (A+) | o  Digital Communication | 4/4 (A) |
| o  Multimedia Systems | 4/4 (A+) | o  Wave Propagation | 4/4 (A) |
| o  Digital Signal Processing | 4/4 (A) | o  Information Theory & Coding | 4/4 (A+) |
| o  Computer Programming | 4/4 (A) | | (M.Sc Course – Audited) |

COMPUTER SKILLS

Programming Languages & Frameworks

- o Python
 - Pytorch
 - Tensorflow
 - Numpy
 - OpenCV
- o C/C++
 - o Matlab
 - o VHDL
 - o Git
 - o L^AT_EX

IDEs/Tools

- o VSCode
- o Simulink
- o CodeVisionAVR
- o Arduino
- o Raspberry Pie
- o Altium:PCD Designer

- o Webots Simulator
- o Isis Proteus Professional
- o 3D High Frequency Simulation Software (HFSS)
- o Advance Design System
- o Xilinx ISE Designer

TEACHING EXPERIENCES

- o Teaching Assistant
 -  Visual Computing Lab Fall 2026
 - Instructor: Prof. Ali Mahdavi-Amiri
 -  Filter Synthesis Winter 2025
 - Instructor: Prof. Zahra Seifi
 -  Logical Circuits Winter 2025
 - Instructor: Prof. Zahra Sadat Shariatmadar

-  Electromagnetics
 - Instructor: Prof. Amir Nader Askarpour
-  Communication Systems
 - Instructor: Prof. Yaser Norouzi

Fall 2024

Winter 2025

REFERENCES

-  Dr. Gh. Moradi, Professor, Electrical Engineering, Amirkabir University of Technology
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-  Dr. F. Abdollahi, Associate Professor, Electrical Engineering, Amirkabir University of Technology
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-  Dr. AN. Askarpour, Associate Professor, Electrical Engineering, Amirkabir University of Technology
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